Final Report: Cryptocurrency Liquidity Predictor

1. Executive Summary

The goal of this project was to build a Machine Learning system capable of predicting the **future liquidity** (trading volume) of cryptocurrencies. Using a **Random Forest Regressor**, the model analyzed historical market metrics to forecast the next day's trading activity. The final model successfully identifies key drivers of liquidity, though its predictive power suggests that external factors (like market sentiment or news) also play a significant role.

2. Methodology

The project followed a standard Data Science pipeline:

- **Data Sources:** Aggregated daily snapshots from CoinGecko (March 16-17, 2022).
- **Preprocessing:** Data was deduplicated and missing values were imputed using symbol-specific means. All numerical features were normalized using a `StandardScaler` to ensure fair comparison between large-cap (Bitcoin) and small-cap coins.
- **Feature Engineering:** A "lagged" target variable was created by shifting the `24h_volume` column backwards by one day. This framed the problem as: *"Given today's metrics, what will the volume be tomorrow?"*.
- **Modeling:** A Random Forest Regressor was trained on 80% of the data and tested on the remaining 20%.

3. Model Performance

The model was evaluated using standard regression metrics on unseen test data:

- **R^2 Score (0.39):** The model explains approximately **39% of the variance** in future liquidity. This is a moderate result, typical for financial data where volatility is high and external "shocks" (such as news or regulation) are frequent.
- **RMSE (1.25):** The Root Mean Squared Error is 1.25 (in scaled units). This represents the average deviation of the model's predictions from the actual values.
- **Bias:** Residual analysis showed a normal distribution of errors centered near zero, indicating the model is not systematically over-predicting or under-predicting.

4. Key Insights & Findings

Analysis of the model's decision-making process revealed interesting patterns about the crypto market:

A. "Volume Begets Volume"

The single most important predictor for *tomorrow's* liquidity was **today's liquidity** (`24h_volume`). It accounted for roughly **58%** of the model's decision-making power. This confirms that trading activity is "sticky"—active coins tend to stay active.

B. Size Matters

Market Capitalization (`mkt_cap`) was the second most critical feature, contributing **37%** to the model's importance. Larger, established coins behave more predictably than smaller ones.

C. Price Action is Irrelevant

Surprisingly, short-term price changes (`1h`, `24h`, `7d` % change) had almost **zero impact** on predicting future volume (combined importance < 2%). A coin pumping or dumping today does not necessarily guarantee higher trading volume tomorrow compared to its baseline.

5. Conclusion

The **Cryptocurrency Liquidity Predictor** successfully established a baseline for forecasting market activity. While it correctly identifies that current volume and market size are the primary drivers of future liquidity, the moderate accuracy (R^2 = 0.39) suggests that market dynamics are complex.

Future Recommendation: To improve performance beyond this baseline, the model would likely need **external data sources** such as social media sentiment (Twitter/X, Reddit), Google Trends data, or macroeconomic indicators, as pure price/volume technicals only tell part of the story.
